Supplementary Material for CFFM-Net: Reliability-Gated Mamba Fusion of Visible and Thermal Images for Detection, Tracking and Multi-Camera Analytics

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This document holds the material moved out of the main paper to meet the page limit. Every number comes from the same result files as the main paper and no claim differs. Tables are numbered S1 onward.

S1. Per-seed results

Table S1 lists every run behind the headline comparison, with the mean and sample standard deviation over seeds. Table S2 gives the per-seed mAP margin of v2 over the concat baseline, which is the quantity averaged in Table I of the main paper.

S2. Fine size bands on M3FD

Table S3 gives the finer size bands behind the small-object analysis of the main paper.

S3. Degradation probe of the round-one models

Table S4 gives the LLVIP degradation probe that preceded the fault benchmark. It uses the round-one models at 30 epochs and one seed.

S4. Ablation detail

Table S5 lists every variant of the ablation of the main paper and Table S6 the margin over the control in each round.

S5. Multi-camera results

Table S7 reports detection, tracking and counting. Zero-shot, the detector has high precision (92.1 %) and low recall (46.5 %), giving a MODA of 42.5 %. These are the numbers of a detector that has never seen the viewpoint: it finds 443 people and misses 509. Fine-tuning on the training frames lifts MODA to 77.4 %, with recall 87.3 % and precision 89.8 %, and cuts the misses to 121 at the price of more false positives (94 against 38). Localisation quality, MODP, is similar for the two, 77.6 % and 76.5 %.

On the ground-plane tracks the system reaches MOTA 79.4 % and IDF1 84.9 %, with 18 identity switches, and tracks 25 of 41 people for most of their presence. Counting is accurate enough to be useful: the mean absolute error of the total count is 1.43 people on a mean of 23.8 people per frame, a relative error of 6.0 %, and the error per zone ranges from 0.43 to 1.18 people. Crossings of the reference line number 11 in one direction and 0 in the other. The busiest zone is the second, with 46 tracks and a mean dwell of 12.4 s, against 5.7 s in the first and 4.0 s in the third.

These results describe the multi-camera layer on RGB input; they say nothing about visible-thermal fusion.

S6. Stream monitor timing

Table S8 gives the detection and recovery delay of the stream monitor and its false alarms for each scripted link fault of the live runtime experiment, with the monitor feeding the model (on) and only observing (off).

S7. Live runtime measurements

Table S9 gives the soak test and the link-fault run of the live runtime experiment on replayed KAIST sequences.

S8. Size bands

Table S10 gives average precision by object size for the headline comparison.

S9. Cost of the evaluated models

Table S11 gives parameters, FLOPs and latency of the evaluated models.

S10. Fault margins

Figure S1 plots the margin of v2 over the concat baseline for each fault of the benchmark in the main paper.

S11. System diagrams

Figure S2 shows the complete system, and Figures S3 to S6 show each of its four parts with the evaluation applied to it. The diagrams carry no results; every number is in the tables of the main paper and of this document.

S12. Result figures

Figures S7 to S16 plot the measurements behind the tables. Each one is drawn directly from the result files of the repository by a single script, and none of them adds a result that the tables do not hold.

TABLE S1
PER-SEED ACCURACY IN PERCENT, 80 EPOCHS. THE LAST THREE COLUMNS ARE COCO SIZE BANDS.

Dataset	Model	Seed	mAP	mAP_{50}	Precision	Recall	Small	Medium	Large
LLVIP	Concat, modality dropout	0	64.8	97.2	96.8	91.7	8.0	64.7	66.1
		1	66.3	96.6	97.0	91.7	9.6	65.8	69.3
		2	65.2	96.6	96.7	90.6	11.0	64.6	68.3
	mean ± std		65.5 ± 0.8	96.8 ± 0.4	96.9 ± 0.2	91.3 ± 0.6	9.6 ± 1.5	65.0 ± 0.6	67.9 ± 1.6
	CFFM-Net v2	0	65.6	97.0	96.4	92.4	8.2	65.5	66.8
		1	66.0	96.6	97.1	91.2	9.0	65.5	68.7
		2	67.1	97.2	96.4	92.4	10.2	66.5	70.5
	mean ± std		66.2 ± 0.8	96.9 ± 0.3	96.6 ± 0.5	92.0 ± 0.7	9.1 ± 1.0	65.8 ± 0.6	68.7 ± 1.9
M3FD	Concat, modality dropout	0	53.5	82.2	85.5	76.4	34.9	67.5	87.0
		1	53.5	81.7	86.4	74.2	34.6	66.9	88.4
		2	52.9	81.8	87.6	74.3	34.8	65.9	86.2
	mean ± std		53.3 ± 0.3	81.9 ± 0.3	86.5 ± 1.1	74.9 ± 1.2	34.8 ± 0.2	66.8 ± 0.8	87.2 ± 1.1
	CFFM-Net v2	0	53.7	82.7	88.8	74.4	34.9	66.8	87.4
		1	54.0	83.4	88.7	75.0	35.1	66.5	87.8
		2	53.9	82.2	85.7	75.0	35.5	66.9	88.0
	mean ± std		53.9 ± 0.1	82.7 ± 0.6	87.7 ± 1.8	74.8 ± 0.4	35.2 ± 0.3	66.7 ± 0.2	87.7 ± 0.3

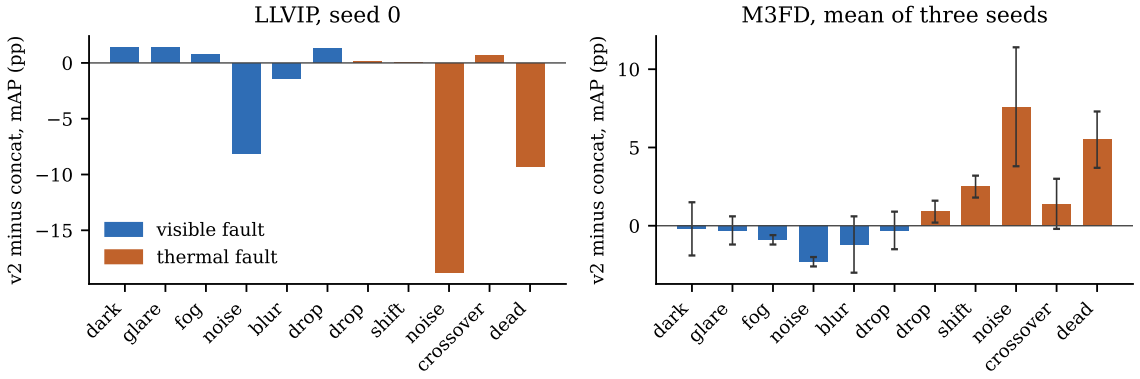


Fig. S1. Margin of CFFM-Net v2 over the concat baseline for each fault at severity three, in mAP points, from Table II of the main paper. Positive means v2 is ahead. LLVIP is one seed; M3FD shows the mean over three seeds with the sample standard deviation as an error bar.

TABLE S2
PER-SEED MAP MARGIN OF CFFM-NET V2 OVER THE CONCAT BASELINE, IN PERCENTAGE POINTS.

Dataset	Seed 0	Seed 1	Seed 2
LLVIP	+0.7	−0.4	+1.9
M3FD	+0.3	+0.5	+1.0

TABLE S3
M3FD AVERAGE PRECISION IN THE FINER SIZE BANDS, PERCENT, PER SEED AND AS MEAN AND SAMPLE STANDARD DEVIATION.

Model	Seed	Very tiny	Tiny	Small
Concat, modality dropout	0	7.5	24.7	48.6
	1	7.8	25.5	47.3
	2	7.6	21.6	49.2
	mean ± std	7.6 ± 0.2	23.9 ± 2.1	48.4 ± 1.0
CFFM-Net v2	0	7.6	24.7	47.9
	1	7.7	24.9	48.4
	2	8.9	25.3	48.6
	mean ± std	8.0 ± 0.8	24.9 ± 0.3	48.3 ± 0.4

TABLE S4
DEGRADATION PROBE ON LLVIP, MAP IN PERCENT, ROUND-ONE MODELS (30 EPOCHS, SEED 0).

Condition	v1	v1, no gate	Concat
Clean	62.9	62.6	65.0
Visible dark	59.9	54.2	61.0
Visible drop	56.7	57.0	61.8
Visible drop, flagged	59.5	57.0	61.8
Thermal drop	2.2	0.7	0.3
Thermal drop, flagged	17.4	0.7	0.3
Thermal shift 4 px	53.3	51.8	51.6
Thermal shift 8 px	32.6	31.1	29.7
Thermal shift 16 px	2.8	2.9	1.7

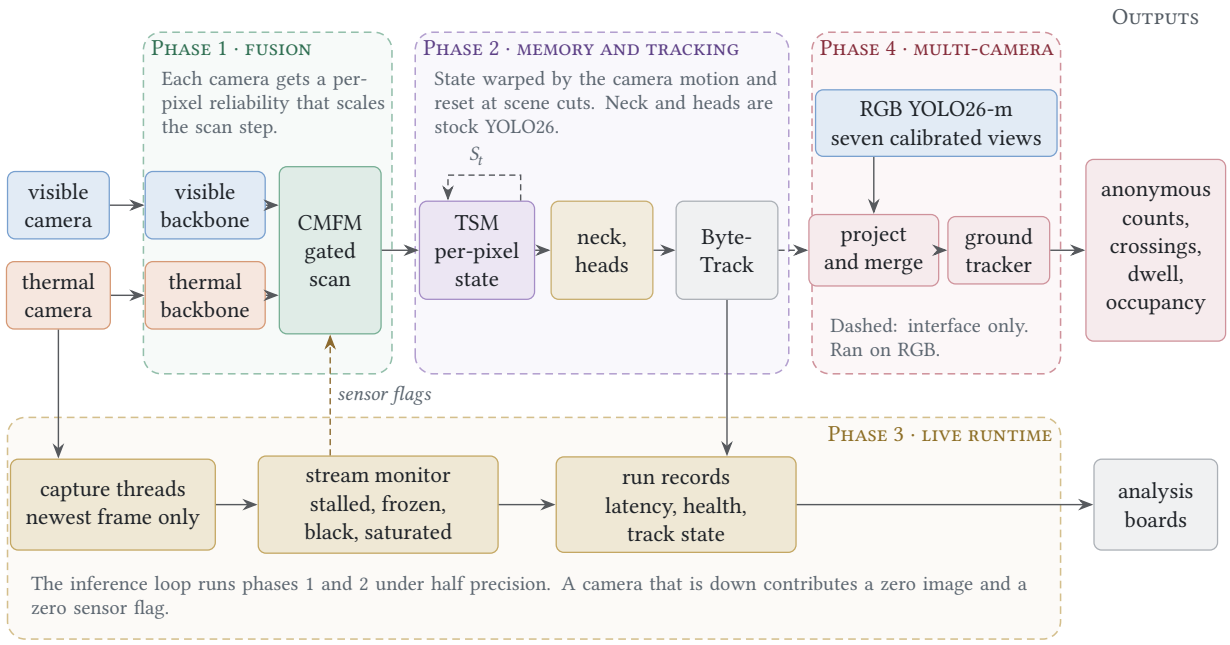


Fig. S2. The complete system. Paired visible and thermal cameras pass through the reliability-gated fusion detector, the Temporal State Memory and ByteTrack, inside a live runtime whose stream monitor raises sensor flags. The multi-camera ground-plane layer was run on an RGB-only detector, so its link from the tracker is the interface only.

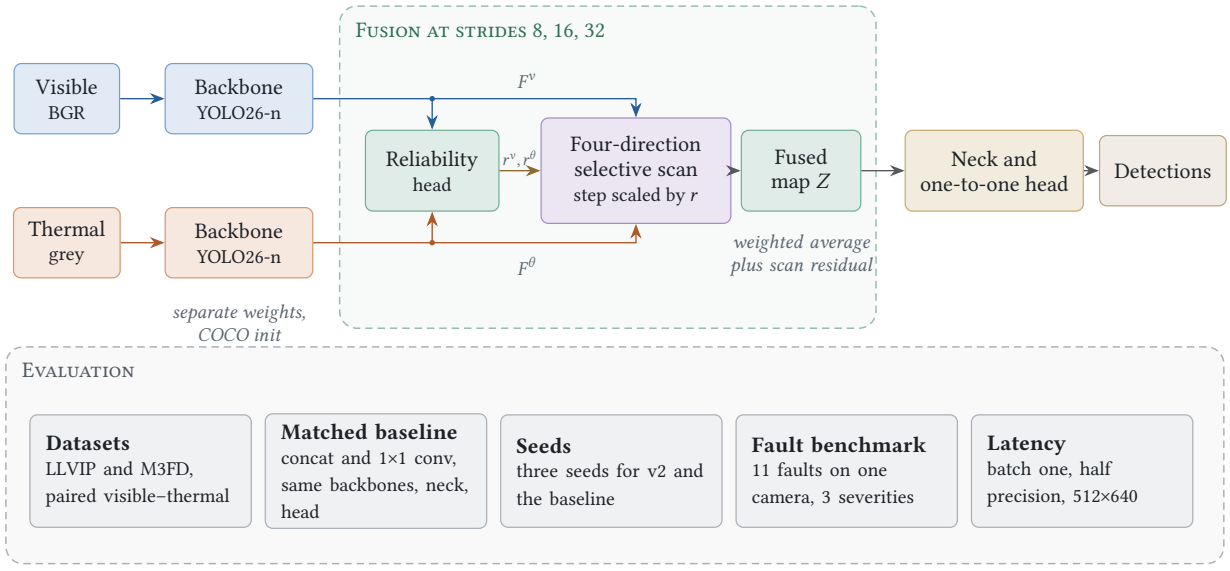


Fig. S3. The fusion detector on still image pairs. Two YOLO26-n backbones feed a fusion block at strides 8, 16 and 32, in which a reliability head scales the step of a four-direction selective scan. The strip below lists the evaluation of the main paper.

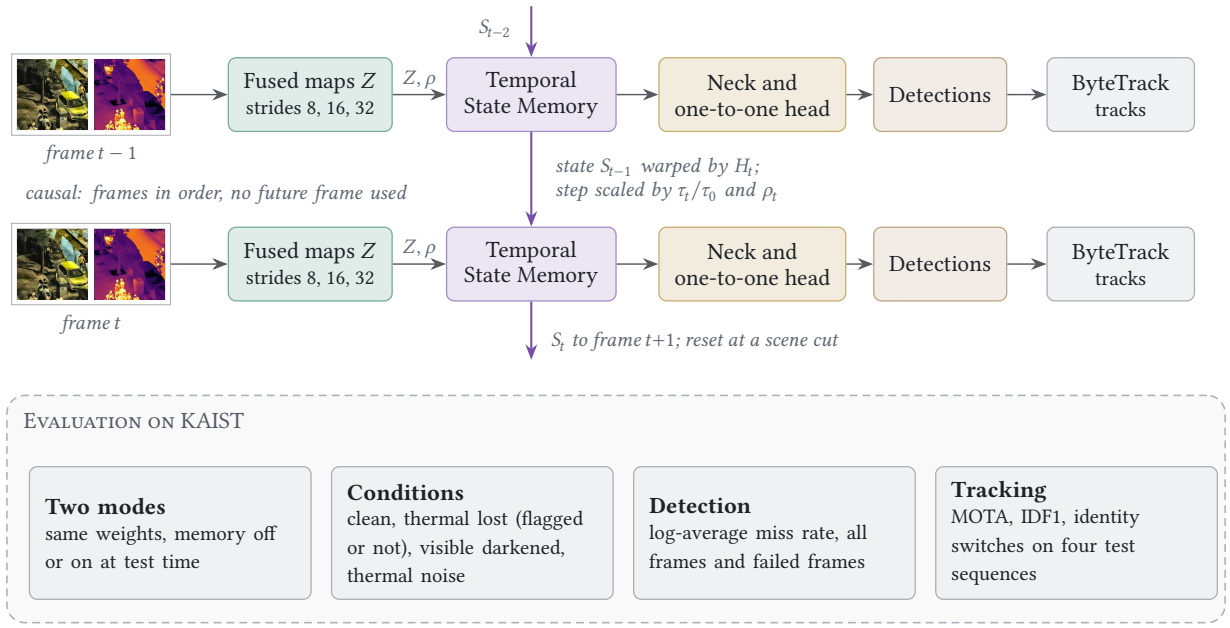


Fig. S4. Video, the Temporal State Memory and tracking. The fused features of each frame update a per-pixel state carried from frame $t-1$ to frame t, warped by the homography and gated by elapsed time and reliability. Detections feed ByteTrack, and KAIST is scored with the memory off and on.

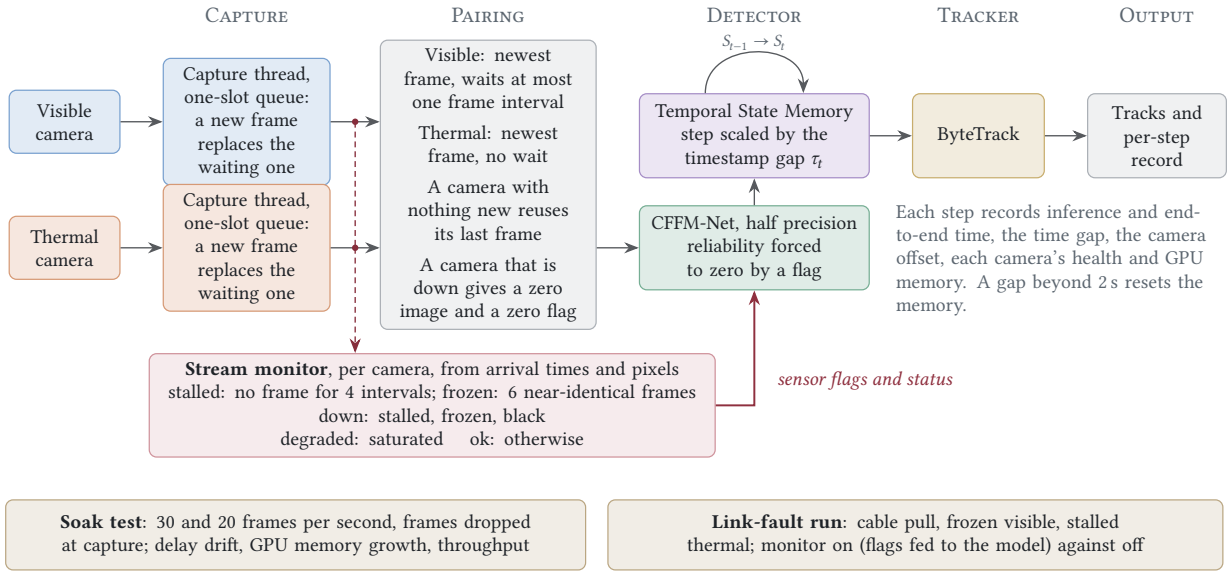


Fig. S5. The live runtime. Two threaded captures with one-slot queues feed a newest-frame pairing step, then the detector with its memory and ByteTrack. A stream monitor watches each camera and feeds sensor flags back to the detector.

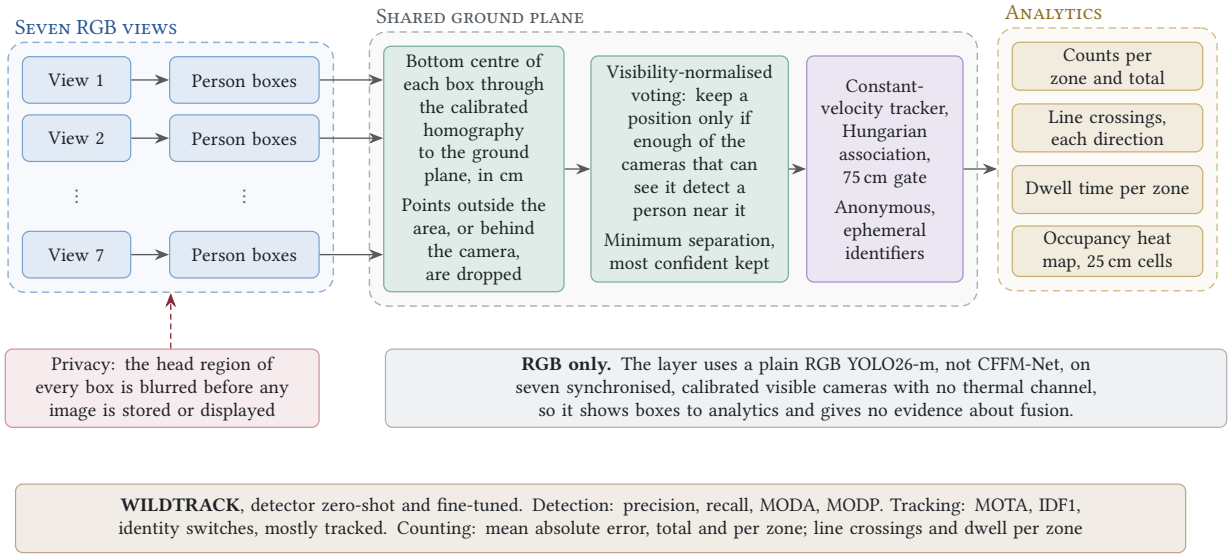


Fig. S6. The multi-camera layer. Person boxes from each RGB view are projected to a shared ground plane, merged across cameras and tracked anonymously, giving counts, line crossings, dwell time and an occupancy map on WILDTRACK. The layer is RGB-only and gives no evidence about fusion.

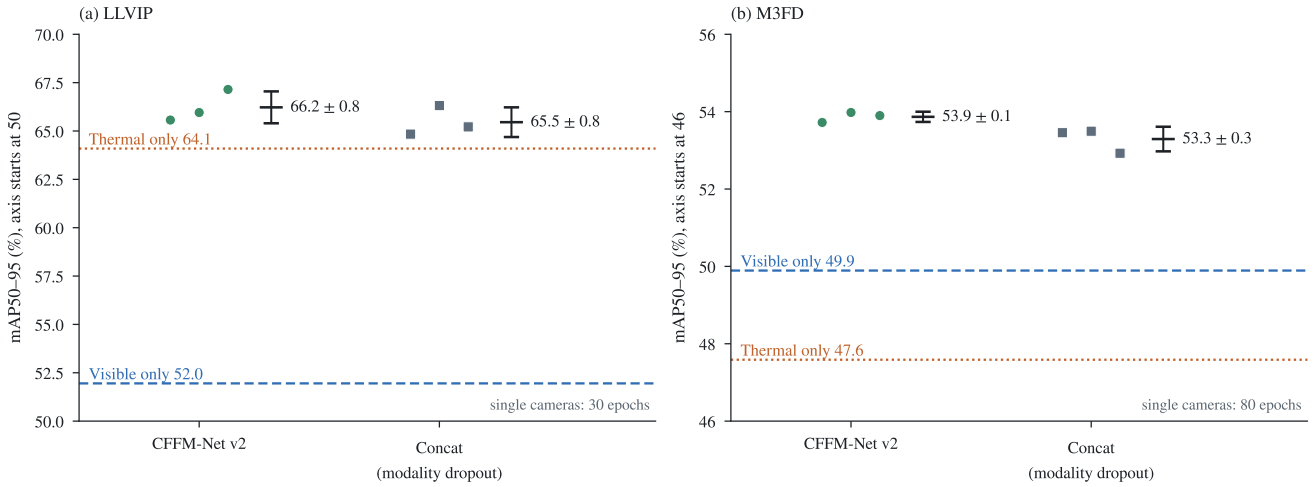


Fig. S7. $mAP_{50\text{-}95}$ of CFFM-Net v2 and the concatenation baseline with modality dropout on LLVIP and M3FD. Dots are the three seeds and the bars are the mean and sample standard deviation. The dashed and dotted lines are the single-camera YOLO26-n baselines at seed 0, trained for 30 epochs on LLVIP and for a matched 80 epochs on M3FD. The vertical axes do not start at zero.

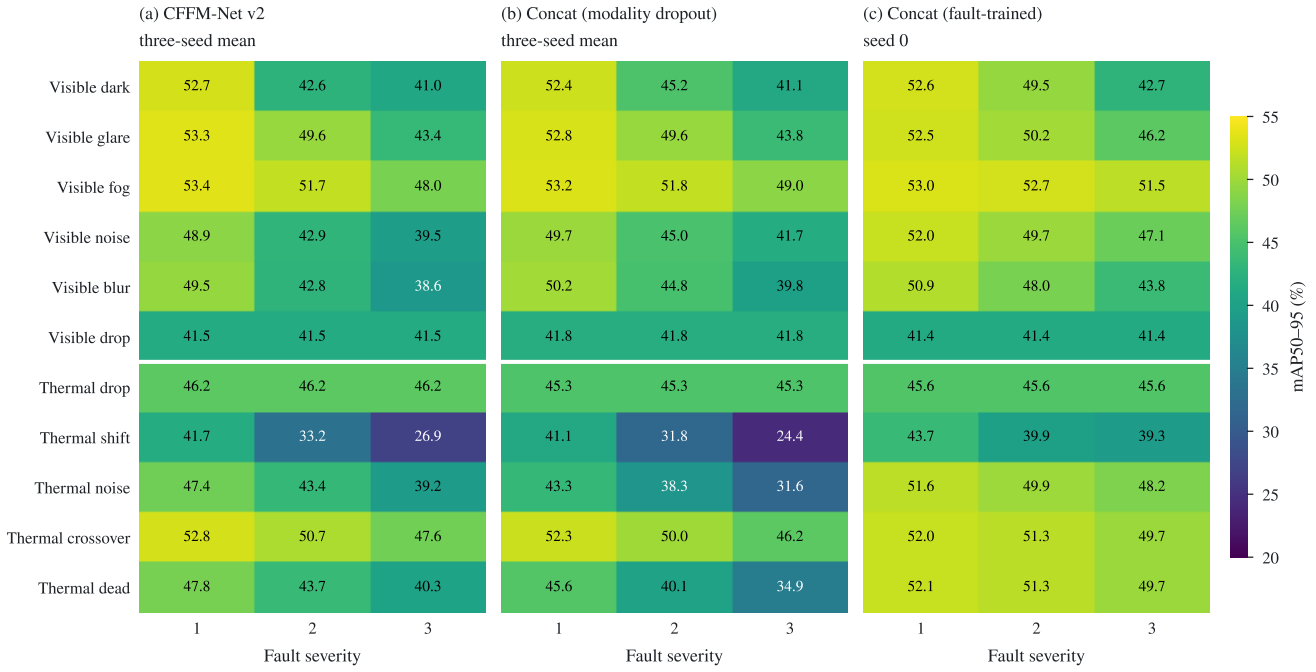


Fig. S8. M3FD $mAP_{50\text{-}95}$ under eleven sensor faults at three severities. CFFM-Net v2 and the concatenation baseline with modality dropout are means over three seeds, and the fault-trained concatenation baseline is seed 0. Dropping a camera has no severity, so those rows are constant.

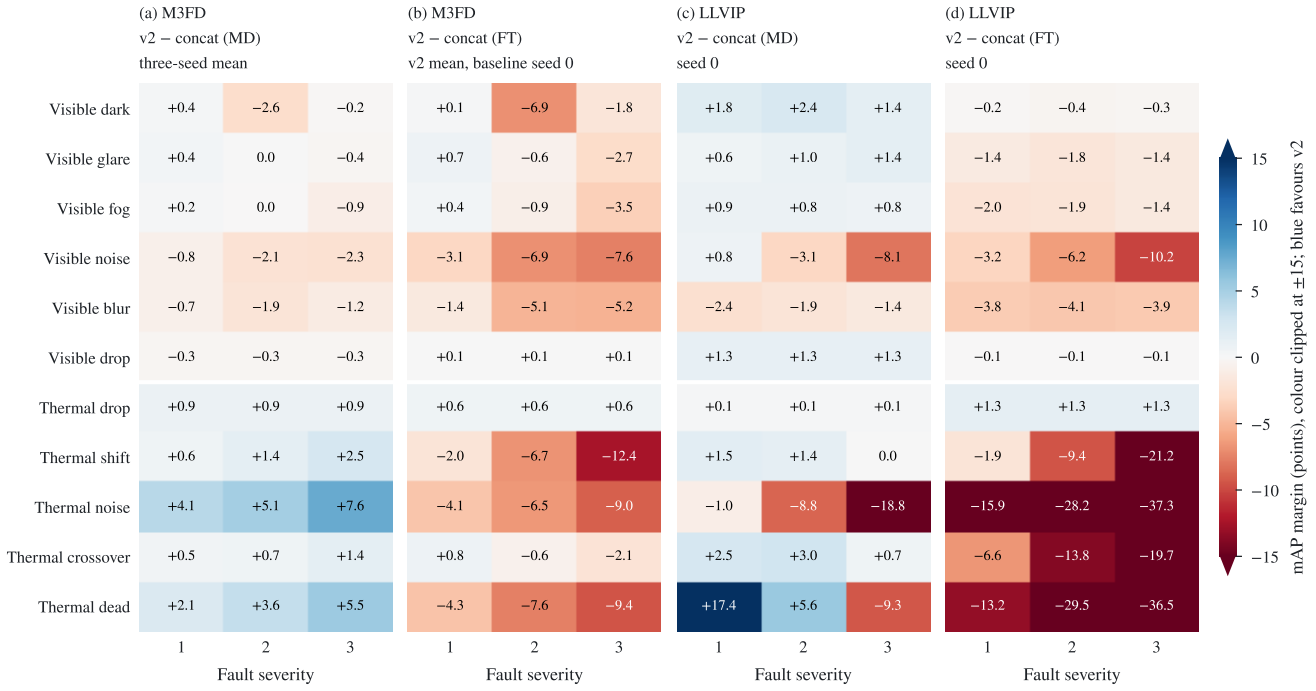


Fig. S9. Margin of CFFM-Net v2 over the two concatenation baselines in mAP points, per fault and severity. Blue cells favour v2 and red cells favour the baseline. On M3FD the margin over the modality-dropout baseline is a three-seed mean and the margin over the fault-trained baseline compares the v2 mean with seed 0, and LLVIP is seed 0 throughout. The colour scale is clipped at 15 points, and every cell prints its value.

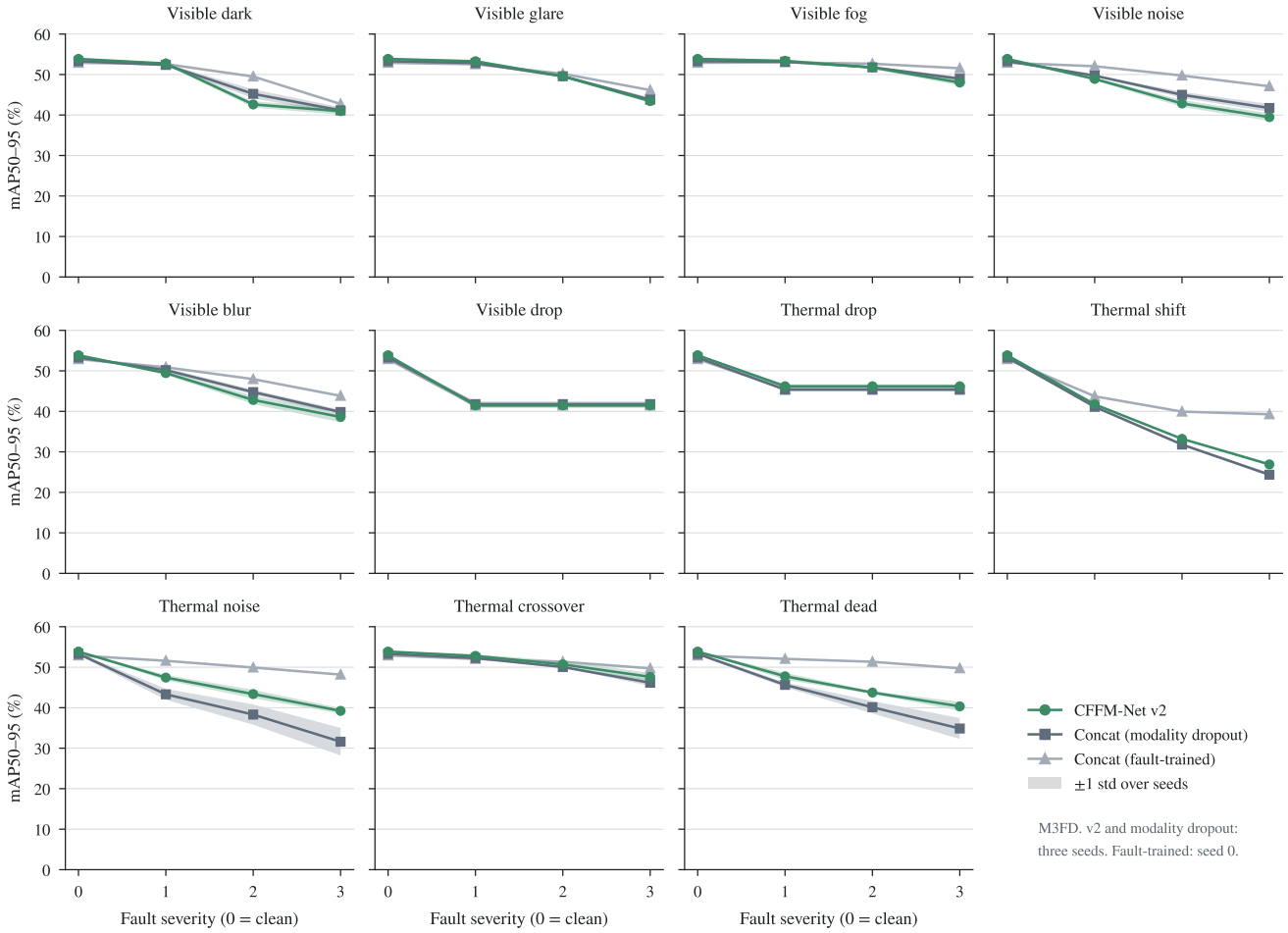


Fig. S10. M3FD $mAP_{50\text{-}95}$ against fault severity for each of the eleven faults. Lines for CFFM-Net v2 and the concatenation baseline with modality dropout are three-seed means with a band of one standard deviation, and the fault-trained baseline is seed 0.

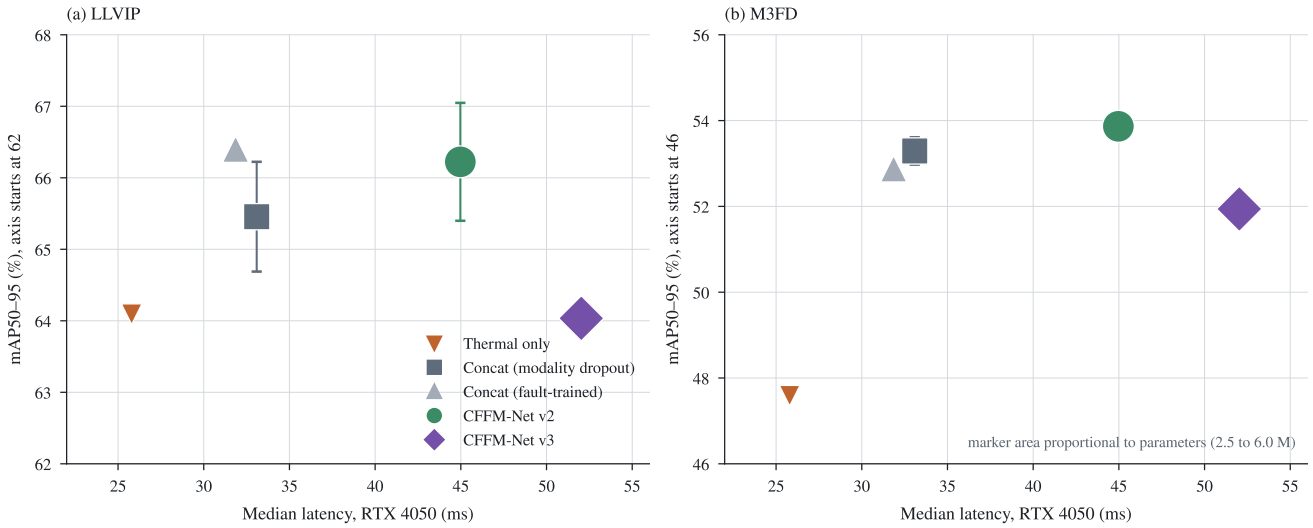


Fig. S11. $mAP_{50\text{-}95}$ against median latency on an RTX 4050 Laptop GPU (FP16, batch 1, 640×512). Marker area is proportional to the parameter count. CFFM-Net v2 and the modality-dropout baseline are three-seed means with one standard deviation, and the other models are seed 0. The thermal-only model on LLVIP was trained for 30 epochs, and the vertical axes do not start at zero.

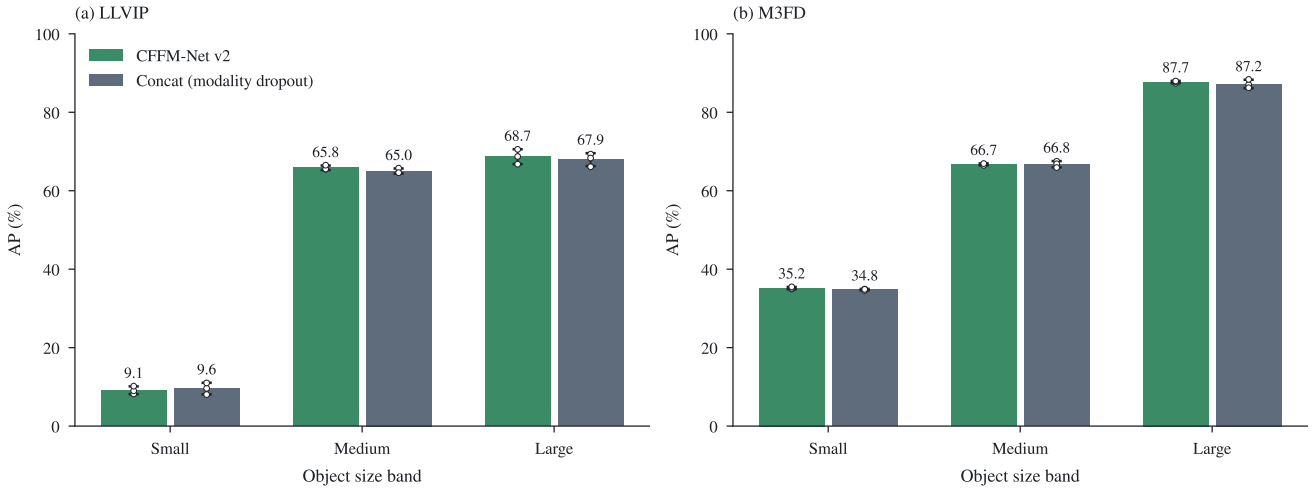


Fig. S12. COCO small, medium and large AP of CFFM-Net v2 and the concatenation baseline with modality dropout. Bars are three-seed means with one standard deviation and the dots are the seeds.

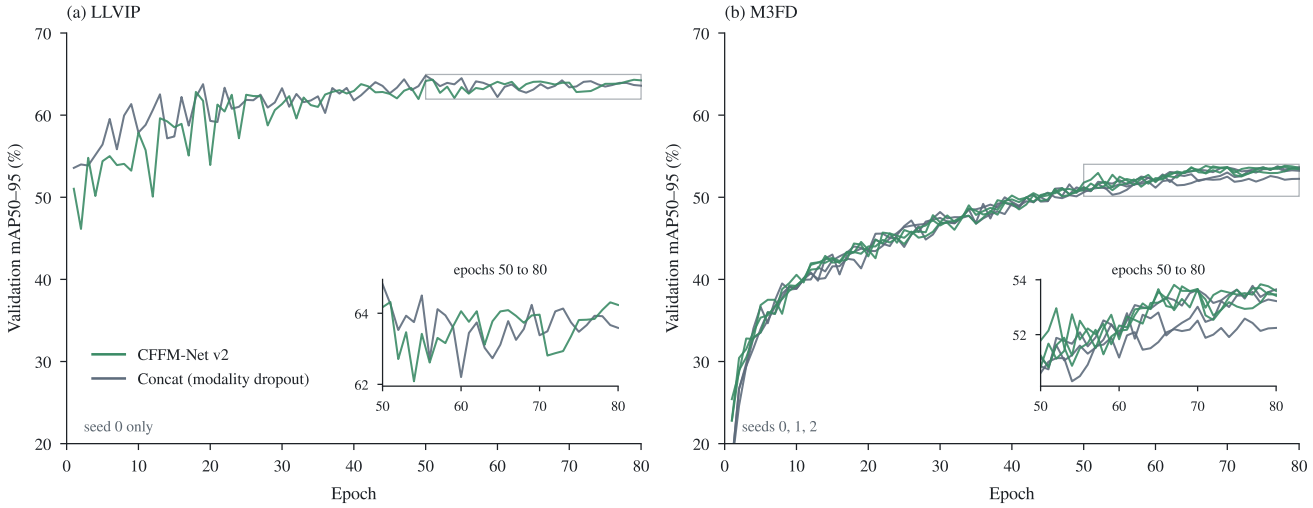


Fig. S13. Validation $mAP_{50\text{-}95}$ against epoch for CFFM-Net v2 and the concatenation baseline with modality dropout, with epochs 50 to 80 enlarged. LLVIP shows seed 0, the only seed whose training log was kept, and M3FD shows seeds 0 to 2.

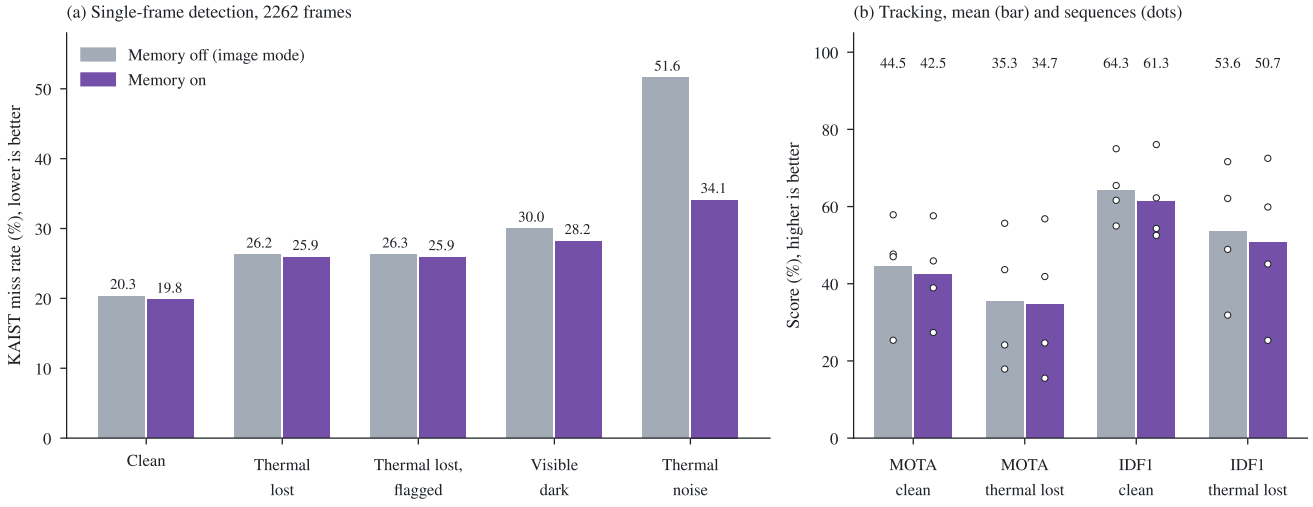


Fig. S14. The Temporal State Memory on KAIST. (a) Miss rate over the 2262 validation frames with the memory off and on under each condition, where lower is better. (b) Tracking MOTA and IDF1 as the mean over four sequences, with one dot per sequence.

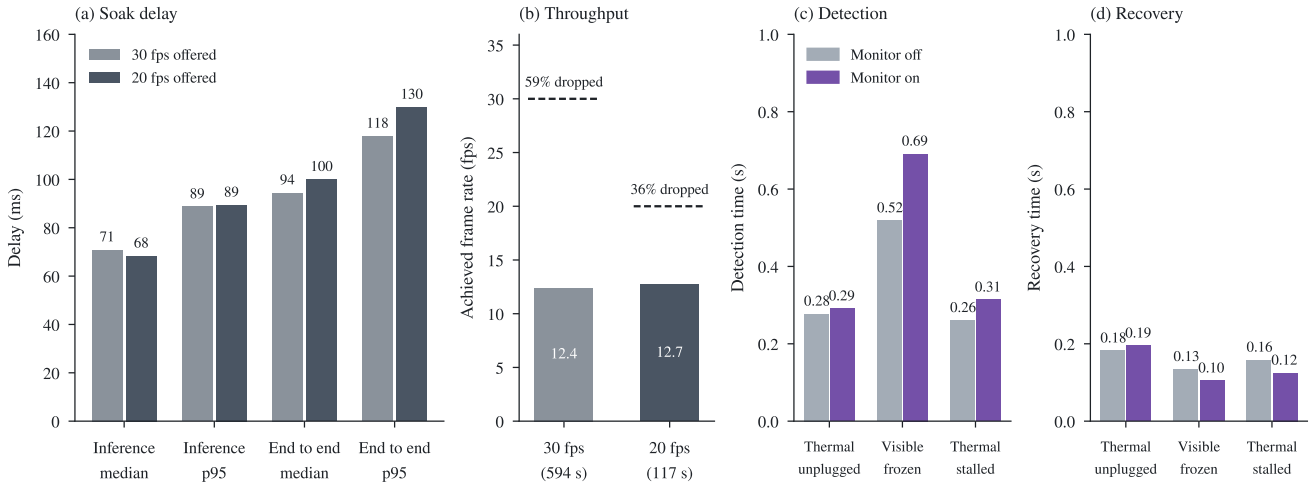


Fig. S15. The live runtime. (a) Median and 95th-percentile inference and end-to-end delay in the soak runs at 30 and 20 offered frames per second. (b) Achieved frame rate against the offered rate, with the share of frames dropped at capture. (c), (d) Time to detect and to recover from three scripted link faults with the stream monitor off and on, one run each.

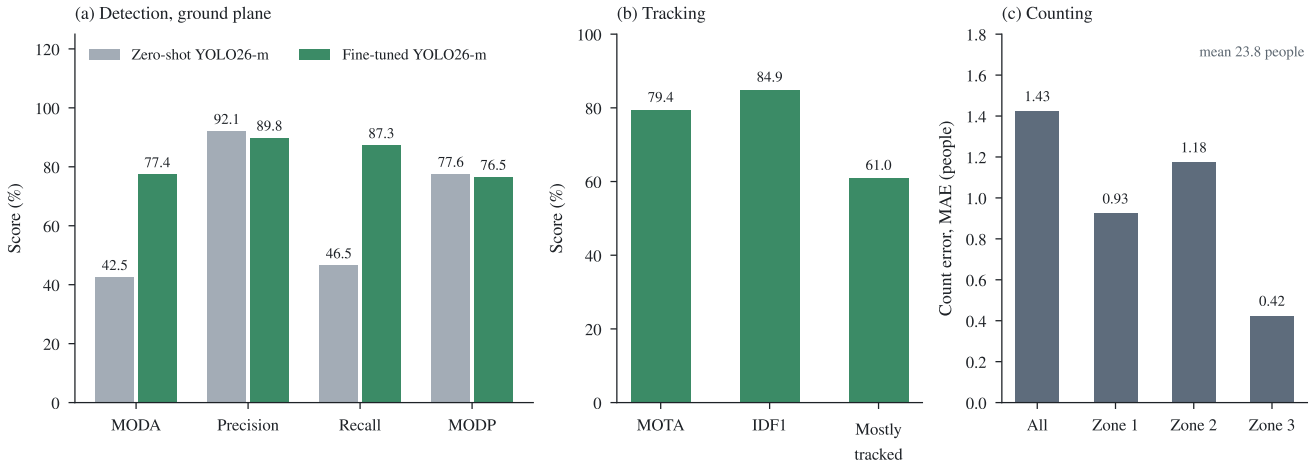


Fig. S16. The multi-camera layer on WILDTRACK with an RGB YOLO26-m detector. (a) Ground-plane detection, zero-shot against fine-tuned. (b) Tracking. (c) Mean absolute count error over the whole ground plane and in three zones.

TABLE S5
ABLATION OF MAP IN PERCENT. A DASH MARKS A RUN THAT WAS NOT MADE.

Variant	LLVIP	M3FD
Round one, 30 epochs, seed 0		
Two-stream concat	64.9	50.5
CFFM-Net v1	62.9	47.9
v1, no gate	62.6	–
v1, gated conv	62.8	–
Concat + P2 head	63.2	–
v1, no P2 head	63.5	–
Rounds three and four, 80 epochs, seed 0		
Concat, modality dropout	64.8	53.5
CFFM-Net v2	65.6	53.7
Concat, fault training	66.4	52.9
CFFM-Net v3	64.0	51.9
v3, no reliability branch	67.2	53.3

TABLE S6
ROUND HISTORY: MAP MARGIN OF CFFM-NET OVER ITS CONTROL IN PERCENTAGE POINTS.

Comparison	LLVIP	M3FD
Round one, v1 against concat (30 epochs)	−2.1	−2.6
Head control, v1 without P2 against concat with P2	+0.3	–
Round three, v2 against concat (seed 0)	+0.7	+0.3
Round three, v2 against concat (three-seed mean)	+0.8	+0.6
Round four, v3 against fault-trained concat	−2.4	−0.9
Round four, v3 against v3 without reliability branch	−3.2	−1.4
Round four, v3 against v2	−1.5	−1.8

TABLE S7
MULTI-CAMERA ANALYTICS. DETECTION USES ZERO-SHOT AND FINE-TUNED YOLO26-M WITH GROUND-PLANE FUSION; TRACKING AND COUNTING ARE REPORTED ONCE.

Measure	Zero-shot	Fine-tuned
Detection		
MODA (%)	42.5	77.4
MODP (%)	77.6	76.5
Precision (%)	92.1	89.8
Recall (%)	46.5	87.3
True positives	443	831
False positives	38	94
False negatives	509	121
Tracking		
MOTA (%)	79.4	
IDF1 (%)	84.9	
ID switches	18	
Mostly tracked of 41	25	
Counting		
Count MAE (people)	1.43	
Relative error (%)	6.0	
Zone one MAE	0.93	
Zone two MAE	1.18	
Zone three MAE	0.43	

TABLE S8
STREAM MONITOR TIMING FOR EACH SCRIPTED FAULT, WITH THE MONITOR FEEDING THE MODEL (ON) AND ONLY OBSERVING (OFF). DELAYS ARE SECONDS FROM ONSET OR END OF THE FAULT TO THE FIRST CHANGE OF STATE.

	Detect (s)		Recover (s)		False alarms (%)	
Fault	On	Off	On	Off	On	Off
Thermal unplugged, 6 s	0.29	0.28	0.19	0.18	4.3	5.0
Visible frozen, 6 s	0.69	0.52	0.10	0.13	0.0	0.0
Thermal stalled, 3 s	0.31	0.26	0.12	0.16	6.1	8.0

TABLE S9
LIVE RUNTIME ON REPLAYED KAIST SEQUENCES. SOAK: PAIRED CAMERAS AT THE STATED RATE. LINK FAULTS: 20 FPS WITH A SCRIPTED CABLE PULL, FREEZE AND STALL. DRIFT IS THE LATE MINUS THE EARLY MEDIAN END-TO-END DELAY. ONE RUN PER ROW.

	Soak		Link faults	
Measure	30 fps	20 fps	Monitor on	Monitor off
Run time (s)	594	117	107	107
Throughput (frames/s)	12.4	12.7	11.8	12.0
Capture drop (%)	58.8	36.3	–	–
End-to-end median (ms)	94.4	100.0	101.5	100.7
End-to-end p95 (ms)	117.9	129.9	139.4	139.4
Inference median (ms)	70.9	68.3	72.6	70.5
Inference p95 (ms)	88.7	89.3	98.6	99.9
Drift (ms)	+0.3	+8.2	+10.7	−2.8
GPU memory growth (MB)	−1	0	+1	−1
Miss rate, all frames (%)	–	–	27.9	27.1
Camera reconnects	–	–	10	10

TABLE S10
AVERAGE PRECISION BY OBJECT SIZE IN PERCENT, MEAN AND SAMPLE STD OVER THREE SEEDS, 80 EPOCHS.

Model	Small	Medium	Large
LLVIP			
Concat, modality dropout	9.6 ± 1.5	65.0 ± 0.6	67.9 ± 1.6
CFFM-Net v2	9.1 ± 1.0	65.8 ± 0.6	68.7 ± 1.9
Margin of v2 (pp)	−0.4 ± 0.5	+0.8 ± 1.1	+0.8 ± 1.4
M3FD			
Concat, modality dropout	34.8 ± 0.2	66.8 ± 0.8	87.2 ± 1.1
CFFM-Net v2	35.2 ± 0.3	66.7 ± 0.2	87.7 ± 0.3
Margin of v2 (pp)	+0.4 ± 0.3	0.0 ± 0.9	+0.5 ± 1.2

TABLE S11
PARAMETERS (MILLIONS), GFLOPS AND LATENCY (MS) ON A RTX 4050 LAPTOP GPU, HALF PRECISION, BATCH ONE, 640×512 INPUT, MEDIAN AND 95TH PERCENTILE PER IMAGE OVER 300 TIMED ITERATIONS AFTER 50 WARM-UP ITERATIONS, ON AN OTHERWISE IDLE GPU.

Model	Params	GFLOPs	Median	p95
YOLO26-n thermal only	2.50	5.9	25.8	47.5
Concat, modality dropout	4.07	9.7	33.1	42.5
Concat, fault training	4.07	9.7	31.9	41.3
CFFM-Net v2	6.02	16.5	45.0	54.7
v3, no reliability branch	6.02	16.5	50.6	65.6
CFFM-Net v3	6.02	16.5	52.0	66.3
v2 over concat, ratio	1.48	1.70	1.36	1.29